

Question number	Answer	Additional guidance	Mark
8(a)(i)	An explanation that includes the following points: • explanation of why another method is not accurate (1) • explanation of why entomology is accurate (1)	e.g. drop in body temperature is dependent on ambient temperature e.g. knowing life cycles and timings insects {colonise dead body in specific order / show succession}	(2)

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8(a)(ii)	An explanation that includes the following points: • some {insects /species} only found in certain habitats (1) • therefore if non-native species found on the body, it has been moved (1)		(2)

Question number	Answer	Additional guidance	Mark
8(b)(i)	An answer that includes the following points: • to prevent animals from {eating / moving} it (1)	ACCEPT scavengers / carnivores appropriate named animal e.g. vulture, wolf IGNORE predators	(1)

Question number	Answer	Additional guidance	Mark
8(b)(ii)	- values read from the graph at 5 and 15 days / a value divided by 240 (1) 0.08 (1)	e.g. of calculation 21 to 22 and 3 / in the range of 0.075 to 0.079166 Bald answer of 0.08 = 2 marks Bald answer between 0.075 and 0.079167 rounded up to more than 2 dps = 1 mark	(2)

Question number	Answer	Additional guidance	Mark
8(b)(iii)	An explanation that includes four of the following points: • {very little / no} change at start because {body not colonised by insects / only insect eggs present} (1) • {small decrease at /end of bloated stage / 3.5 days} due to wild animals eating flesh (1) • {large decrease in mass / at beginning of active stage / 5 days} as holes in flesh allow gases to escape (1) • {decrease in mass / during active and advanced stages} as insects eat the flesh (1) • some mass remains because bones cannot be {eaten / digested} (1)	ACCEPT very few microorganisms present decomposition has not started ACCEPT decomposition by microorganisms small volume of gas released some insects have started to eat loss of water ACCEPT mass remains constant because bones remain remains cannot be digested NB If no marks are awarded allow 1 mark for a description of decomposition	(4)

Question number	Answer	Additional guidance	Mark
8(b)(iv)	to identify the <u>species</u> (that laid the eggs) / to determine time until hatching to find out when they were laid (1)		(1)

Question number	Answer	Additional guidance	Mark
8(b)(v)	An explanation that includes the following points: - succession means that the species of insects (found on the decomposing mammal) will {change with time / change with stage (of decomposition) / appear in a specific sequence} (1) <i>Lucilia</i> appears on the body first (1) - then {<i>Cochliomyia</i> / <i>Chrysomya</i>} appear on the body and <i>Lucilia</i> {numbers decrease / is out competed} (1) - then <i>Ophyra</i> arrives and {<i>Cochliomyia</i> / <i>Chrysomya</i>} {decrease in number / are out competed} (1)	ACCEPT pioneer species	(3)

